Normal Distribution

1 Basic discussion

I want to build up a $f(x) \propto ke^{bx^2}$ which should be symmetric, uni-modal.

$$\begin{aligned}
&\Rightarrow \quad f(x) = ke^{-bx^2}, x\varepsilon\mathbb{R} \text{ if } x = -\infty, f(x) \to 0 \\
&\therefore \quad \int_{-\infty}^{\infty} f(x)dx = 1 \\
&\Rightarrow \quad \int_{-\infty}^{\infty} ke^{-bx^2}dx = 1 \\
&\Rightarrow \quad 2k\int_0^{\infty} e^{-bx^2}dx = 1 \\
&\Rightarrow \quad 2k\int_0^{\infty} e^{-z}\frac{dz}{2b\sqrt{\frac{z}{b}}} = 1 \\
&\Rightarrow \quad \frac{k}{\sqrt{b}}\int_0^{\infty} e^{-z}z^{\frac{1}{2}-1}dz = 1 \\
&\Rightarrow \quad \frac{k}{\sqrt{b}}\sqrt{1/2} = 1 \Rightarrow \frac{k}{\sqrt{b}}\sqrt{\pi} = 1 \Rightarrow k = \sqrt{\frac{b}{\pi}}
\end{aligned}$$

Let's try to interpret the constants by first moment.

$$\mu = E(X) = \int_{-\infty}^{\infty} xf_X(x)dx = \int_{-\infty}^{\infty} xke^{-bx^2}dx = 0 \text{ [odd function]}.$$

The first moment is unable to capture k and b. So, we will go to the second moment.

$$\begin{aligned}
E\left(X^2\right) = \int_{-\infty}^{\infty} x^2 f_X(x)dx &= k\int_{-\infty}^{\infty} x^2 e^{-bx^2}dx \\
&= 2k\int_0^{\infty} x^2 e^{-bx^2}dx = 2k\int_0^{\infty} \frac{z}{b}e^{-z}\frac{dz}{2b\sqrt{\frac{z}{b}}} \\
&= \frac{k}{b\sqrt{b}}\int_0^{\infty} e^{-z}z^{\frac{3}{2}-1}dz \\
&= \frac{k}{b\sqrt{b}}\Gamma 3/2 = \frac{k}{b\sqrt{b}}\frac{1}{2}\Gamma\frac{1}{2} \\
&= \frac{k}{b\sqrt{b}}\frac{1}{2}\sqrt{\pi}
\end{aligned}$$

$E(X) = 0 \Rightarrow \text{Var}(X) = E\left(X^2\right) - E^2(X) \Rightarrow \quad V(X) = E\left(X^2\right) = \sigma^2$
So,

$$\sigma^2 = \frac{k}{2b\sqrt{b}}\sqrt{\pi} \quad \left[k = \sqrt{\frac{b}{\pi}}\right] \Rightarrow b = \frac{1}{2\sigma^2}, k = \frac{1}{\sigma\sqrt{2\pi}}$$

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2\sigma^2}x^2}, \quad x\varepsilon\mathbb{R}$$

We tried to make our distribution symmetric wrt 0, lets convert that thing to a. $\quad Y = X + a$.
PDF of Y, $f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\frac{(y-a)^2}{\sigma^2}}, \quad y \in \mathbb{R}$.
Then, we again need to interpret a.

$$E(Y-a) = \int_{-\infty}^{\infty}(y-a)\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{y-a}{\sigma}\right)^2} = \int_{-\infty}^{\infty} z\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{z}{\sigma}\right)^2} = 0$$

$$\Rightarrow E(Y) = \mu = a$$

$$\therefore \quad f_Y(y) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{y-\mu}{\sigma}\right)^2}$$

1. Show that $f(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty$ is a PDF.

$f(x) > 0$, it is obvious.

$$\int_{-\infty}^{\infty}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 dx} = \frac{2}{\sigma\sqrt{2\pi}}\int_0 e^{-frac12u}\sigma dt$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}}\int_0^{\infty}\frac{1}{\sqrt{2u}}e^{-u}du$$

$$= \frac{\sqrt{2}}{\sqrt{\pi}}\frac{1}{\sqrt{2}}\int_0^{\infty}u^{-1/2}e^{-u}du$$

$$\frac{1}{\sqrt{pi}}\sqrt{\pi} = 1$$

Notation: If X has the PDF $f_X(x) = \frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, \quad x \in \mathcal{R}, \mu \in \mathbb{R}$
Then we write, $X \sim N\left(\mu, \sigma^2\right)$

2 Moments

Odd order central moments are 0.

$$\begin{aligned}\mu_{2r-1} &= \int_{-\infty}^{\infty}(x-\mu)^{2r-1}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx && \frac{x-\mu}{\sigma} = z\\ &= \int_{-\infty}^{\infty}(\sigma z)^{2r-1}\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\frac{z^2}{\sigma^2}}\sigma dz && \Rightarrow dx = \sigma dz\\ &= 0(\text{As the integrant is an odd function.})\end{aligned}$$

$$
\begin{aligned}
\mu_{2r} &= \int_{-\infty}^{\infty} (x-\mu)^{2r} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
&= \int_{-\infty}^{\infty} (z\sigma)^{2r} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{z^2}{2}} \sigma dz \\
&= 2\sigma^{2r} \int_0^{\infty} z^{2r} \frac{e^{-z^2/2}}{\sqrt{2\pi}} dz \\
&= 2\sigma^{2r} \int_0^{\infty} (\sqrt{2u})^{2r} \frac{e^{-u}}{\sqrt{2\pi}} \frac{du}{\sqrt{2u}} \\
&= \frac{\sigma^{2r}}{\sqrt{\pi}} \int_0^{\infty} (2u)^r \frac{e^{-u}}{\sqrt{u}} du \\
&= \frac{2^r \sigma^{2r}}{\sqrt{\pi}} \int_0^{\infty} u^{r-\frac{1}{2}} e^{-u} du \\
&= \frac{2^r \sigma^{2r}}{\sqrt{\pi}} \sqrt{r+\frac{1}{2}} \\
&= \frac{2^r \sigma^{2r}}{\sqrt{u}} \left(r-\frac{1}{2}\right)\left(r-\frac{3}{2}\right)\left(r-\frac{5}{2}\right)\ldots\Gamma(1/2) \\
&= 2^r \sigma^{2r} \frac{(2r-1)}{1} \frac{(2r-3)}{2} \frac{(2r-5)}{2} \cdots \frac{5}{2}\frac{3}{2} \\
&= \sigma^{2r} 1.3 \cdot 5 \cdot \cdots \cdot (2r-3)(2r-1)
\end{aligned}
$$

Here $\gamma_1 = \frac{\mu_3}{\mu_2^{3/2}} = 0$ and $\beta_2 = \frac{\mu_4}{\mu_2^2} = \frac{3\sigma^4}{(\sigma^2)^2} = 3$
Therefore, we take $\Gamma_2 = \beta_2 - 3$ as a measure of excess of kurtosis.

3 Median

Note that $f_X(\mu - h) = \frac{1}{\sigma\sqrt{2\pi}} e^{-h^2}\sigma^2 = f_X(\mu + h)$.
$\Rightarrow f_X(x)$ is symmetric about μ.
$\Rightarrow N\left(\mu, \sigma^2\right)$ distribution is symmetric about μ. Hence μ is the median of the distribution.

4 Mode

$f_X(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$ is maximum iff $\frac{1}{2}\left(\frac{x-u}{\sigma}\right)^2$ is minimum.

$$
\begin{aligned}
&\text{iff } \left(\frac{x-\mu}{\sigma}\right)^2 = 0 \\
&\text{iff } \quad x = \mu
\end{aligned}
$$

5 Mean Deviation about μ

$$
\begin{aligned}
E|X-\mu| &= \int_{-\infty}^{\infty} |x-\mu| \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \qquad z = \frac{x-\mu}{\sigma} \\
&= \int_{-\infty}^{\infty} |z\sigma| \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}z^2} \sigma dz \qquad\qquad \sigma dz = dx \\
&= \frac{2}{\sqrt{2\pi}} \sigma \int_0^{\infty} z e^{-\frac{1}{2}z^2} dz \\
&= \sigma\sqrt{\frac{2}{\pi}} \int_0^{\infty} e^{-u} du \\
&= \sigma\sqrt{\frac{2}{\pi}} \left[\frac{e^{-u}}{-1}\right]_0^{\infty} \\
&= \sigma\sqrt{\frac{2}{\pi}}
\end{aligned}
$$

For a Normal distribution,

$$\frac{\text{MD about mean}}{\text{sd}} = \sqrt{\frac{2}{\pi}} \Rightarrow \text{MD} = \sqrt{\frac{2}{\pi}}\sigma \simeq 0.8\sigma \Rightarrow \text{MD} = \frac{4}{5}\text{SD}$$

(This is known as Garry's Ratio)

1. If $X \sim N\left(\mu, \sigma^2\right)$. Find $E|X-\mu|$ and $\text{Var}(|X-\mu|)$.

$$E|X-\mu| = \sigma\sqrt{\frac{2}{\pi}}$$

$\text{Var}[|X-\mu|] = E(X-\mu)^2 - E^2|X-\mu| = \sigma^2 - \sigma^2\frac{2}{\pi} = \sigma^2\left(1-\frac{2}{\pi}\right)$

2. If $X \sim N\left(\mu, \sigma^2\right)$ for any $\sigma\varepsilon\mathbb{R}$. Show that, $E|X-a| \geqslant \sigma\sqrt{\frac{2}{\pi}}$

$$E|X-a| \geqslant E|X-\mu| = \sigma\sqrt{\frac{2}{\pi}} \text{ (MD about median is the least need a proof}$$

3. X_1, X_2, X_3, X_4 be independent $N\left(0, k^2\right)$ RV, $k = 1, 2, 3, 4$ respectively. Find the probability that exactly 2 of these variable are less than 0.

Z = Number of X are less than zero. $Z \sim \text{Bin}\left(4, \frac{1}{2}\right)$.
As, $p\left(X_i < 0\right) = \frac{1}{2}, p(Z=2) = \binom{4}{2}\left(\frac{1}{2}\right)^4$.

4. Find the mean and variance of the distribution whose PDF is $f_X(x) = ce^{-\frac{1}{24}\left(x^2-6x+9\right)}, x \in \mathbb{R}$.

$$\Rightarrow -\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2 = -\frac{1}{24}(x-3)^2 \Rightarrow \mu = 3 \Rightarrow \sigma = \sqrt{12} = 2\sqrt{3}$$

Mean = 3, variance = 12.

5. Let $X \sim N\left(\mu, \sigma^2\right)$. What would be the distribution of X if $\sigma^2 = 0$?

$\sigma^2 = 0, P(X = m) = 1$. It is a degenerate distribution.

6 Standard Normal Distribution

If $X \sim N\left(\mu = 0, \sigma^2 = 1\right)$, its PDF will be given by $f_X(x) = \frac{1}{\sqrt{2\pi}}e^{-\frac{1}{2}x^2}$ and the distribution is known as Standard Normal Distribution.

The standard Normal PDF is denoted by,

$$\phi(t) = \frac{1}{\sqrt{2\pi}}e^{-z^2/2}, z \in \mathbb{R}$$

It's CDF is given by $F(x) = \int_{-\infty}^{x}\phi(t)dt = \int_{-\infty}^{x}\frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt = \Phi(x)$.

$\Phi(x)$ can not be evaluated exactly but $\Phi(x)$ can be computed numerically. The value of $\Phi(x)$ for different values of x is provided in Biometrika Vol I.

7 Properties

1. Here $\phi(x)$ is symmetric about 0.$\Rightarrow$ mean = median = mode = 0.
   Note that, $\phi(x) = \frac{1}{\sqrt{2\pi}}e^{-x^2/2}$ decreases rapidly to zero as $|x|$ increases.

2. $\Phi(k) + \Phi(-k) = 1$

   $\Phi(k) = \int_{-\infty}^{k} \phi(t)dt =$ The area bounded by the curve below the line $y = k$.

   $= p(X > -k) = 1 - \Phi(-k)$ (Due to symmetry)

   $\Rightarrow \Phi(k) + \Phi(-k) = 1$

   Implication: If $\Phi(k), k > 0$ is known, we can easily find out the value of $\Phi(-k)$ using the fact $\Phi(k) + \Phi(-k) = 1$. Therefore, it is only necessary to provide the table of $\Phi(k)$ for positive values of k and it reduces fact of the computational labour.

3. If $X \sim N(0, 1)$. Find the distribution of $Y = |X|$. Hence, find $E \mid x|$ and $V(\mid X \mid)$.

   $\rightarrow$ CDF of $Y = |X|$ is, $F_Y(y) = P(Y \leqslant y) = P(|X| \leqslant y)$

$$= \begin{cases} p[-y \leqslant X \leqslant y] & y \geqslant 0 \\ 0 & y < 0 \end{cases}$$

$$= \begin{cases} \Phi(y) - \Phi(-y) & \text{if } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \Phi(y) - (1 - \Phi(y)) & \text{if } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases} = \begin{cases} 2\Phi(y) - 1 & \text{if } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

   PDF of Y will be, $\frac{d}{dy}F_Y(y) = \begin{cases} 2\phi(y) & \text{if } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$

$$= \begin{cases} 2\frac{1}{\sqrt{2\pi}}e^{-y^2/2} & \text{if } y \geqslant 0 \\ 0 & \text{otherwise} \end{cases}$$

   This distribution is called folded normal distribution.

$$\begin{aligned} E(|X|) &= \int_0^\infty |x| 2\tfrac{1}{\sqrt{2\pi}} e^{-x^2/2} dx = \sqrt{\tfrac{2}{\pi}} \\ V(|X|) &= E\left(X^2\right) - E^2(X) = 1 - \tfrac{2}{\pi} \\ E\left(X^2\right) &= \int_0^\infty x^2 2\tfrac{1}{\sqrt{2\pi}} e^{-x^2/2} dx \\ &= \sqrt{\tfrac{2}{\pi}} \int_0^\infty 2ze^{-z} \tfrac{dz}{\sqrt{2z}} = \tfrac{2}{\sqrt{\pi}} \int_0^\infty z^{-1+\frac{3}{2}} e^{-z} dz = \tfrac{2}{\sqrt{\pi}} \tfrac{1}{2} \sqrt{\pi} = 1 \end{aligned}$$

$$\begin{aligned} \tfrac{x^2}{2} &= z \\ \tfrac{2xdx}{2} &= dz \end{aligned}$$

8 Point of Inflexion

$\rightarrow$ $X \sim N(\mu, \sigma^2)$

$$\begin{aligned} f(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty \\ f'(x) &= \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} \left(-\frac{1}{2}\right) 2\left(\frac{x-\mu}{\sigma}\right) = -\left(\frac{x-\mu}{\sigma^2}\right) f(x) \\ f''(x) &= \frac{-1}{\sigma^2} f(x) + \left(\frac{x-\mu}{\sigma^2}\right)^2 f(x) \\ &= \frac{f(x)}{\sigma^2}\left[\frac{(x-\mu)^2}{\sigma^2} - 1\right] \\ \therefore \quad f''(x) = 0 &\Rightarrow \frac{(x-\mu)^2}{\sigma^2} = 1 \quad \Rightarrow x - \mu = \sigma \\ &\Rightarrow x = \pm\sigma + \mu = \mu \pm \sigma \end{aligned}$$

9 Area of Normal Distribution

$$P(Z \leqslant k) = \int_{-\infty}^{k} \phi(z)dz = \Phi(k)$$
$$P(Z \leqslant 0) = \int_{-\infty}^{0} \phi(z)dz = \Phi(0) = \frac{1}{2}$$
$$P(Z \geqslant k) = 1 - P(Z \leqslant k) = 1 - \Phi(k)$$
$$P(Z \leqslant -k) = \int_{-\infty}^{-k} \phi(z)dz = \Phi(-k) = 1 - \Phi(k)$$
$$P\left(k_1 \leqslant Z \leqslant k_2\right) = P\left(Z \leqslant k_2\right) - P\left(Z \leqslant k_1\right) = P\left(k_2\right) - \Phi\left(k_1\right)$$
$$P\left(-k_1 \leqslant Z \leqslant k_2\right) = P\left(Z \leqslant k_2\right) - P\left(Z \leqslant -k_1\right) = P\left(k_2\right) - \Phi\left(-k_1\right) = \Phi(k_1) + \Phi(k_2) - 1$$

10 Quartile Deviation

Q_1 and Q_3 be the first and third quantile of the distribution.

$$\begin{array}{ll} \int_{-\infty}^{Q_1} f(x)dx = 0.25 & \int_{-\infty}^{Q_3} f(x)dx = 0.75 \\ \Rightarrow \int_{-\infty}^{Q_1} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0.25 & \Rightarrow \int_{-\infty}^{Q_3} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx = 0.75 \\ \Rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{Q_1-\mu}{\sigma}} e^{-\frac{1}{2}z^2} dz = 0.25 & \Rightarrow \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{Q_3-\mu}{\sigma}} e^{-\frac{1}{2}z^2} dz = 0.75 \\ \Rightarrow \int_{-\infty}^{Q_1-\mu} \phi(z)dz = 0.25 & \Rightarrow \int_{-\infty}^{Q_3-\mu} \phi(z)dz = 0.75 \\ \Rightarrow \Phi\left(\frac{Q_1-\mu}{\sigma}\right) = 0.25 & \Rightarrow \Phi\left(\frac{Q_3-\mu}{\sigma}\right) = 0.75 \end{array}$$

From table,

$$\Phi(-0.67) = 0.25 \Rightarrow Q_1 = w - 0.670 \text{ and } Q_3 = \mu_1 + 0.670$$

So,

$$\frac{Q_3 - Q_1}{2} = 0.670. \Rightarrow QD = 0.670$$
$$QD = \frac{2}{3}SD$$
$$\therefore SD : MD : QD = \sigma : \frac{4}{5}\sigma : \frac{2}{3}\sigma = 16 : 12 : 10$$

5. Show that $\left(\frac{1}{x} - \frac{1}{x^3}\right)\phi(x) \leqslant 1 - \Phi(x) \leqslant \frac{1}{x}\phi(x), x > 0$. $\lim_{x\to\infty} \frac{x(1-\Phi(x))}{\phi(x)} = 1$, for $x > 0$

$$\begin{aligned} 1 - \Phi(x) &= \int_x^\infty \phi(t)dt = \int_x^\infty \frac{1}{t}t\phi(t)dt \\ &= \frac{1}{t} - \phi(t)|_x^\infty - \int_x^\infty -\frac{1}{t^2}(-\phi(t))dt \\ &= \frac{\phi(x)}{x} - \int_x^\infty \frac{1}{t^3}t\phi(t)dt \\ &= \frac{\phi(x)}{x} - \left[\frac{1}{t^3} - \phi(t)|_x^\infty - \int_x^\infty \frac{-3}{t^4}(-\phi(t))dt\ \right] \\ &= \frac{\phi(x)}{x} - \frac{\phi(x)}{x^3} + 3\int_x^\infty \frac{\phi(t)}{t^4}dt \end{aligned}$$

(i) $1 - \Phi(x) = \frac{\phi(x)}{x} - \int_x^\infty \frac{\phi(t)}{t^2}dt \quad \Rightarrow 1 - \Phi(x) \leqslant \frac{\phi(x)}{x}$
(ii) $1 - \Phi(x) = \frac{\phi(x)}{x} - \frac{\phi(x)}{x^3} + 3\int_x^\infty \frac{\phi(t)}{t^4}dt \Rightarrow 1 - \Phi(x) \geqslant \phi(x)\left\{\frac{1}{x} - \frac{1}{x^3}\right\}$

Combining (i) & (ii) we get,

$$\phi(x)\left\{\frac{1}{x}-\frac{1}{x^3}\right\} \leqslant 1-\Phi(x) \leqslant \frac{\phi(x)}{x}$$
$$\Rightarrow \left\{1-\frac{1}{x^2}\right\} \leqslant \frac{x(1-\phi(x))}{\phi(x)} \leqslant 1$$
$$\lim_{x\to\infty}\left\{1-\frac{1}{x^2}\right\} \leqslant \lim_{x\to\infty}\frac{x(1-\phi(x))}{\phi(x)} \leqslant \lim_{x\to\infty} 1$$
$$\therefore \lim_{x\to\infty}\frac{x(1-\phi(x))}{\phi(x)} = 1 \quad [\text{ By sandwich theorem}]$$

For large x ie. $x \approx 3$ or $\geqslant 3$, the values of $\Phi(x)$ are not given in the table. Then $(1-\Phi(x)) \simeq \frac{1}{x}\frac{1}{\sqrt{2t}}e^{-x^2/2}$, the approximate relationship. can be used to compute $\Phi(x)$ for $x > 3$.

6. Let $X \sim N\left(0, 5^2\right)$ .Y be an independent random variable taking the values ± 1 with equal probability. Find the distribution of $S = XY + \frac{X}{Y}$ and $T = XY - \frac{X}{Y}$ and also find the distribution of $\left(\frac{S+T}{10}\right)^2$.

The distribution function of S is,

$$\begin{aligned}
F_S(s) = P(S \leqslant s) &= P(S \leqslant s, Y = 1) + P(S \leqslant s, Y = -1) \\
&= P(S \leqslant s \mid Y = 1)P(Y = 1) + P(S \leqslant s \mid Y = -1)P(Y = -1) \\
&= P\left(X \leqslant \frac{s}{2}\right)\frac{1}{2} + P\left(-X \leqslant \frac{s}{2}\right)\frac{1}{2} \\
&= P\left(X \leqslant \frac{s}{2}\right) (X \& -X \text{ have the same distribution due to symmetry}) \\
&= P\left(\frac{X-0}{5} \leqslant \frac{\frac{s}{2}-0}{5}\right) \\
&= \Phi\left(\frac{s}{10}\right)
\end{aligned}$$

Hence, $S \sim N\left(0, 10^2\right)$

$$T = XY - \frac{X}{Y} = \begin{cases} 0 & Y = +1 \\ 0 & Y = -1 \end{cases} \Rightarrow T = 0 \text{ with prob 1.}$$

T is degenerate at 0.

7. $X \sim N\left(0, \sigma^2\right)$ Let Y be such that,

$$Y = \begin{cases} X & \text{if} |x| \leq 1 \\ -X & \text{if} |x| \geq 1 \end{cases}$$

What is the distribution of Y ?

$$\begin{aligned}
F_Y(y) &= P(Y \leqslant y) \\
&= P(Y \leqslant y, |X| \leqslant 1) + P(Y \leqslant y, |X| > 1) \\
&= P(Y \leqslant y, |X| \leqslant 1)P(|X| \leqslant 1) + P(Y \leqslant y, |X| > 1)P(|X| > 1) \\
&= P(X \leqslant y)P(|X| \leqslant 1) + P(-X \leqslant y)P(|X| > 1) \\
&= P(X \leqslant y)P(|X| \leqslant 1) + P(X \leqslant y)P(|X| > 1) \\
&= P(X \leqslant y)[P(|X| \leqslant 1) + P(|X| > 1)]. \\
&= P(X \leqslant y) \cdot 1 = F_X(y).
\end{aligned}$$

Then $Y = X$. So, $Y \sim N\left(0, \sigma^2\right)$

8. If $X \sim N\left(\mu, \sigma^2\right)$, for any function $g(x)$ such that $g'(x)$ exists. Show that $E[(X-\mu)g(x)] = \sigma^2 E\left[g'(X)\right]$ (provided all the expectations exists).

$$E[(X-\mu)g(x)] = \int_{-\infty}^{\infty}(x-\mu)g(x)\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx$$

Note that,

$$\begin{aligned}
&\int(x-\mu)\frac{1}{\sigma\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}dx \\
&= \int \sigma z\frac{1}{\sigma\sqrt{2\pi}}e^{-z^2/2}\sigma dz \\
&= \frac{\sigma}{\sqrt{2\pi}}\int ze^{-z^2/2}dz \\
&= \frac{\sigma}{\sqrt{2\pi}}\int e^{-p}dp = \frac{\sigma}{\sqrt{2\pi}}\left(-e^{-p}\right) \\
&= -\frac{\sigma}{\sqrt{2\pi}}e^{-\frac{1}{2}\left(\frac{x-u}{\sigma}\right)^2}
\end{aligned}$$

Note:

$$\begin{aligned}
\Rightarrow \mu_{2r} &= \sigma^2(2r-1)\mu_{2r-1} \\
&= \sigma^4(2r-1)(2r-3)\mu_{2r-4} \\
&= \sigma^{2r}(2r-1)(2r-3)\ldots 5.3.1
\end{aligned}$$

9. Let $Z \sim N(0,1)$. Define $f(z) = \int_0^z \phi(t)dt$. Then $\Phi(z)(1-\Phi(z)) = \frac{1}{4} - F^2(z)$

$$\Phi(z) = \int_{-\infty}^{z}\phi(t)dt = \int_{-\infty}^{0}\phi(t)dt + \int_0^z \phi(t)dt = \frac{1}{2} + F(z)$$

$$1-\Phi(z) = 1 - \frac{1}{2} - F(z) = \frac{1}{2} - F(z)$$

So, $\left\{\frac{1}{2} + F(z)\right\}\left\{\frac{1}{2} - F(z)\right\} = \frac{1}{4} - F^2(z)$ [Proved].

10. $X \sim N(0,1)$. Find the correlation coefficient between $X, \Phi(x), \rho_{x,\Phi(x)}$.

$$
\begin{aligned}
E(\Phi(X)) &= \int_{-\infty}^{\infty} \Phi(x)\phi(x)dx \\
&= \int_0^1 w dw = \left.\frac{w^2}{2}\right|_0^1 = \frac{1}{2} \\
E\left[\Phi^2(X)\right] &= \int_0^{\infty} \Phi^2(x)\phi(x)dx \\
&= \int_0^1 \omega^2 d\omega = \frac{\omega^3}{3} = \frac{1}{3} \\
\mathrm{Var}(\Phi(X)) &= \frac{1}{3} - \left(\frac{1}{2}\right)^2 = \frac{1}{12} \\
\mathrm{Cov}(X, \Phi(X)) &= E(X\Phi(X)) - E(X)E(\Phi(X)), \quad E(X) = 0. \\
&= E(X\Phi(X)) \\
&= \int_{-\infty}^{\infty} x\Phi(x)\phi(x)dx \\
&= \Phi(x)(-\Phi(x))]_{-\infty}^{\infty} + \int_{-\infty}^{\infty} \phi(x)\phi(x)dx \\
&= \int_{-\infty}^{\infty} \left[\frac{1}{\sqrt{2\pi}} e^{-x^2/2}\right]^2 dx = \frac{1}{\pi}\int_0^{\infty} e^{-x^2} dx = \frac{1}{2\sqrt{\pi}} \\
\rho_{X,\Phi(X)} &= \frac{\frac{1}{2\sqrt{\pi}}}{\sqrt{\frac{1}{12}}\sqrt{1}} = \sqrt{\frac{3}{\pi}}
\end{aligned}
$$

11. If $X \sim N\left(\mu, \sigma^2\right)$. Find $E(\Phi(X))$.

$$
\begin{aligned}
E(\Phi(X)) &= \int_{-\infty}^{\infty} \Phi(x) f_X(x) dx \\
&= \int_{-\infty}^{\infty} \left\{\int_{-\infty}^{x} \phi(t)dt\right\} \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2} dx \\
&= \int_{-\infty}^{\infty} \left(\int_{-\infty}^{x} N(0,1)dz\right) \quad N\left(\mu, \sigma^2\right) dx \\
[P[T \leqslant x] &= \sum_{x=0}^{\infty} P(T \leqslant x, X = x) \\
&= \sum_x P(T \leqslant x \mid X = x) P(X \leqslant x) \\
&= \sum_x P(T \leqslant x) f_X(x) \\
&= \sum_x \left\{\sum_{t:T\leqslant x} f_T(t)\right\} f_x(x)\Bigg] \\
&= P\left[\frac{(T-x)+\mu}{\sqrt{1+\sigma^2}} \leqslant \frac{\mu}{\sqrt{1+\sigma^2}}\right] \\
&= \Phi\left(\frac{\mu}{\sqrt{1+\sigma^2}}\right)
\end{aligned}
$$

So, $E(\Phi(X)) = \Phi\left(\frac{\mu}{\sqrt{1+\sigma^2}}\right) = \begin{cases} \geq \frac{1}{2} & \mu \geq 0 \\ < \frac{1}{2} & \mu < 0 \\ = 0 & \mu = 0 \end{cases}$

12. For $z > 0$, show that $\dfrac{z\phi(z)}{1+z^2} < 1 - \Phi(z)$.

$1 - \Phi(z) = \int_x^\infty \phi(t)dt = \int_x^\infty \left[\frac{t^2}{1+t^2} + \frac{1}{1+t^2}\right]\phi(t)dt.$

$$= \int_x^\infty \frac{t}{1+t^2}\phi(t)dt + \int_x^\infty \frac{\phi(t)}{1+t^2}dt.$$

13. If $a > 0$. Find $\lim_{x\to\infty} \frac{1-\Phi\left(x+\frac{a}{x}\right)}{1-\Phi(x)}$.

$\lim_{x\to\infty} \frac{1-\Phi\left(x+\frac{a}{x}\right)}{1-\Phi(x)} = \lim_{x\to\infty} -\phi\left(x+\frac{a}{x}\right)\left(1-\frac{a}{x^2}\right) = \lim_{x\to\infty} \frac{\frac{1}{\sqrt{2\pi}}e^{-\frac{\left(x+\frac{a}{x}\right)^2}{2}}\left(\frac{x^2-a}{x^2}\right)}{\frac{1}{\sqrt{2\pi}}e^{-x^2/2}}$

$= \dfrac{e^{-\frac{\left(x^2+\frac{a^2}{x^2}+2a\right)}{2}}\left(\frac{x^2-a}{x^2}\right)}{e^{-x^2/2}} = \dfrac{e^{-x/2}e^{-a^2/2x^2}e^{-a}\left(\frac{x^2-a}{x^2}\right)}{e^{-x/2}} \to e^{-a}.$

14. If $a > 0, x > 0$, show that $\frac{1-\Phi\left(x+\frac{a}{x}\right)}{1-\Phi(x)} < e^{-a}$

$1 - \Phi\left(z+\frac{a}{z}\right) - \int_{z+\frac{a}{z}}^\infty \phi(t)dt = \int_{z+\frac{a}{2}}^\infty \frac{1}{\sqrt{2\pi}}e^{-t^2/2}dt$

$= \frac{1}{2\sqrt{\pi}}\int_{\frac{1}{2}\left(z+\frac{a}{2}\right)^2}^\infty v^{-v/2}dv = \frac{1}{2\sqrt{\pi}}\left\{\left[-k^{-1/2}e^{-k}\right]_{1/2}^\infty \left(z+\frac{a}{2}\right)^2 - \int_{\frac{1}{2}\left(z+\frac{a}{z}\right)^2}^\infty \frac{3}{2}k^{-3/2}e^{-k}dk\right.$

$\leqslant \frac{1}{2\sqrt{\pi}}\dfrac{e^{-\frac{1}{2}\left(z+\frac{a}{z}\right)^2}}{\left(z+\frac{a}{z}\right)} \cdot \sqrt{2} \leqslant e^{-\frac{1}{2}\left(z+\frac{a}{z}\right)^2}$

$1 - \Phi(z) \leqslant e^{-z^2/2}$

$\frac{1-\Phi\left(z+\frac{a}{z}\right)}{1-\Phi(z)} = \frac{e^{-\frac{1}{2}\left(z+\frac{a}{z}\right)^2}}{e^{-2z^2/2}} = e^{-\frac{1}{2}\left(z^z+2a+\frac{a^2}{z^2}-z^z\right)} = e^{-a-\frac{a^2}{2z^2}} = e^{-a}/e^{a^2/2z^2} < e^{-a}.$

15. $\left|\frac{z\Phi(-z)}{\Phi(z)} - 1\right| < \frac{1}{z^2}$

From, $1 - \Phi(z) > \left(\frac{1}{z} - \frac{1}{z^3}\right)\phi(z)$.

or, $z\Phi(-z) > \left(1 - \frac{1}{z^2}\right)\phi(z)$

or, $\frac{z\Phi(-z)}{\phi(z)} > \left(1 - \frac{1}{z^2}\right)$.

or, $\frac{z\Phi(-z)}{\phi(z)} - 1 > -\frac{1}{z^2}$ or, $-\left[\frac{z\Phi(-z)}{\Phi(z)} - 1\right] < \frac{1}{z^2}$

$\therefore \left|\frac{z\Phi(-z)}{\phi(z)} - 1\right| < \frac{1}{z^2}$ (Proved).

16. $p(Z \geqslant a) \leqslant e^{-a^2/2}$

$p(Z \geqslant a) = \int_a^\infty \frac{1}{\sqrt{2\pi}}e^{-\frac{z^2}{2}}dz.$

$= \int_{a^2/2}^\infty \frac{1}{\sqrt{2\pi}}e^{-t}\frac{dt}{\sqrt{2t}} = \frac{1}{2\sqrt{\pi}}\int_{a^2/2}^\infty e^{-t}t^{-1/2}dt$

$= \frac{1}{2\sqrt{\pi}}\left\{\left[-\frac{1}{\sqrt{t}}e^{-t}\right]_{a^2/2}^\infty - \frac{1}{2}\int_{a^2/2}^\infty t^{-3/2}e^{-t}dt\right\}$

$= \frac{1}{2\sqrt{\pi}}e^{-a^2/2}\frac{1}{\sqrt{\frac{a^2}{2}}} - \frac{1}{4\sqrt{\pi}}\int_{a^2/2}^\infty t^{-3/2}e^{-t}dt$

$p(Z \geqslant a) < e^{-a^2/2}\frac{1}{a\sqrt{2\pi}} < e^{-a^2/2}$ (Proved).